Interview-Style Programming Questions: Loops, Strings, and Number Operations

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1. Print Numbers from 1 to n

Question: Write a program to print numbers from 1 to n. **Explanation:** Use a loop starting from 1 to n and print each number. - **Input:** n = 5 - **Output:** 1 2 3 4 5

- # Program 1: Print Numbers from 1 to n

- n = int(input("Enter n: "))
- for i in range(1, n + 1):
- print(i, end=" ")

- OUTPUT:
- `Enter n: 5`
- `1 2 3 4 5`

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2. Print Numbers from m to n

Question: Write a program to print numbers from m to n. **Explanation:** Loop from m to n and print values. - **Input:** m = 3, n = 7 - **Output:** 3 4 5 6 7

- # Program 2: Print Numbers from m to n

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- for i in range(m, n + 1):
- print(i, end=" ")

- OUTPUT:
- Enter m: 3
- Enter n: 7
- 3 4 5 6 7

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3. Print Numbers from n to 1 in Reverse

Question: Write a program to print numbers in reverse from n to 1. **Explanation:** Use a loop starting from n and decrement to 1. - **Input:** n = 5 - **Output:** 5 4 3 2 1

- # Program 3: Print Numbers from n to 1 in Reverse

- n = int(input("Enter n: "))
- for i in range(n, 0, -1):
- print(i, end=" ")

- OUTPUT:
- Enter n: 5
- 5 4 3 2 1

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4. Print Numbers from n to m in Reverse

Question: Write a program to print numbers from n to m in reverse. **Explanation:** Start from n and go down to m. - **Input:** n = 10, m = 6 - **Output:** 10 9 8 7 6

- # Program 4: Print Numbers from n to m in Reverse

- n = int(input("Enter n: "))
- m = int(input("Enter m: "))
- for i in range(n, m - 1, -1):
- print(i, end=" ")

- OUTPUT:
- Enter n: 10
- Enter m: 6
- 10 9 8 7 6

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5. Sum of n Natural Numbers

Question: Write a program to calculate the sum of first n natural numbers. **Explanation:** Use formula or loop to sum from 1 to n. - **Input:** n = 5 - **Output:** 15

- # Program 5: Sum of n Natural Numbers

- n = int(input("Enter n: "))
- sum = 0
- for i in range(1, n + 1):

- sum = sum + i
- print(sum)

- OUTPUT:
- Enter n: 5
- 15

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6. Factorial of a Number

Question: Write a program to find the factorial of a number. **Explanation:** Multiply all numbers from 1 to n. - **Input:** n = 5 - **Output:** 120

- # Program 6: Factorial of a Number

- n = int(input("Enter n: "))
- fact = 1
- for i in range(1, n + 1):
- fact = fact * i
- print(fact)

- OUTPUT:
- Enter n: 5
- 120

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7. Sum of m to n Numbers

Question: Write a program to find the sum of all numbers from m to n. **Explanation:** Loop from m to n and add values. - **Input:** m = 3, n = 6 - **Output:** 18

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- sum = 0
- for i in range(m, n + 1):
- sum = sum + i
- print("Sum =", sum)

- OUTPUT:
- Enter m: 3
- Enter n: 6
- Sum = 18

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8. Product of m to n Numbers

Question: Write a program to find the product of numbers from m to n. **Explanation:** Loop from m to n and multiply values. - **Input:** m = 2, n = 4 - **Output:** 24

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- product = 1
- for i in range(m, n + 1):
- product = product * i
- print("Product =", product)

- OUTPUT:
- Enter m: 2
- Enter n: 4
- Product = 24

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9. Print Factors of a Number

Question: Write a program to print all factors of a given number. **Explanation:** Check divisibility of number from 1 to n. - **Input:** n = 6 - **Output:** 1 2 3 6

- n = int(input("Enter a number: "))
- for i in range(1, n + 1):
- if n % i == 0:
- print(i, end=" ")

- OUTPUT:
- Enter a number: 6
- 1 2 3 6

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10. Count of Factors

Question: Write a program to count how many factors a number has. **Explanation:** Increment count when divisible. - **Input:** n = 6 - **Output:** 4

- n = int(input("Enter a number: "))
- count = 0
- for i in range(1, n + 1):
- if n % i == 0:
- count = count + 1
- print(count)

- OUTPUT:
- Enter a number: 6
- 4

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11. Prime Number Check

Question: Check if a number is prime. **Explanation:** A number is prime if it has exactly 2 factors. - **Input:** n = 7 - **Output:** Prime

- n = int(input("Enter a number: "))
- count = 0
- for i in range(1, n + 1):
- if n % i == 0:
- count = count + 1
- if count == 2:
- print("Prime")
- else:
- print("Not Prime")

- OUTPUT:
- Enter a number: 7
- Prime

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12. Even Numbers from m to n

Question: Print all even numbers between m and n. **Explanation:** Use loop and check if divisible by 2. - **Input:** m = 3, n = 10 - **Output:** 4 6 8 10

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- for i in range(m, n + 1):
- if i % 2 == 0:
- print(i, end=" ")

- OUTPUT:
- Enter m: 3
- Enter n: 10
- 4 6 8 10

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13. Odd Numbers from m to n

Question: Print all odd numbers between m and n. **Explanation:** Check if number % 2 != 0. - **Input:** m = 3, n = 10 - **Output:** 3 5 7 9

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- for i in range(m, n + 1):
- if i % 2 != 0:
- print(i, end=" ")

- OUTPUT:
- Enter m: 3
- Enter n: 10
- 3 5 7 9

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14. Count of Even and Odd Numbers

Question: Count how many even and odd numbers are in the range m to n. **Explanation:** Use counters for even and odd. - **Input:** m = 3, n = 7 - **Output:** Even = 2, Odd = 3

- m = int(input("Enter m: "))
- n = int(input("Enter n: "))
- even = 0
- odd = 0
- for i in range(m, n + 1):
- if i % 2 == 0:
- even = even + 1
- else:
- odd = odd + 1
- print("Even =", even)
- print("Odd =", odd)

- OUTPUT:
- Enter m: 3
- Enter n: 7
- Even = 2
- Odd = 3

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15. Reverse a String

Question: Reverse a given string. **Explanation:** Use slicing or loop. - **Input:** “hello” - **Output:** “olleh”

- string = input("Enter a string: ")
- reverse = string[::-1]
- print(reverse)

- OUTPUT:
- Enter a string: hello
- olleh

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16. Check for Palindrome String

Question: Check if a string is a palindrome. **Explanation:** Compare string with its reverse. - **Input:** “madam” - **Output:** Palindrome

- string = input("Enter a string: ")
- reverse = string[::-1]
- if string == reverse:

- print("Palindrome")
- else:
- print("Not Palindrome")

- OUTPUT:
- Enter a string: madam
- Palindrome

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17. Sum of Digits

Question: Calculate the sum of digits of a number. **Explanation:** Use loop and % 10 to extract digits. - **Input:** 123 - **Output:** 6

- number = int(input("Enter a number: "))
- sum = 0
- while number > 0:
- digit = number % 10
- sum = sum + digit
- number = number // 10
- print(sum)

- OUTPUT:
- Enter a number: 123
- 6

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18. Product of Digits

Question: Calculate the product of digits. **Explanation:** Multiply digits extracted from number. - **Input:** 123 - **Output:** 6

- number = int(input("Enter a number: "))
- product = 1
- while number > 0:
- digit = number % 10
- product = product * digit
- number = number // 10
- print(product)

- OUTPUT:
- Enter a number: 123
- 6

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19. Armstrong Number Check

Question: Check if a number is an Armstrong number. **Explanation:** Sum of cube of digits equals the number. - **Input:** 153 - **Output:** Armstrong number

number = int(input("Enter a number: "))

- temp = number
- sum = 0
- while temp > 0:
- digit = temp % 10
- sum = sum + (digit ** 3)
- temp = temp // 10
- if sum == number:
- print("Armstrong number")
- else:
- print("Not Armstrong number")

- OUTPUT:
- Enter a number: 153
- Armstrong number

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20. Reverse a Number

Question: Reverse the digits of a number. **Explanation:** Use loop with % and // to reverse. - **Input:** 123 - **Output:** 321

- number = int(input("Enter a number: "))
- reverse = 0
- while number > 0:
- digit = number % 10

- reverse = reverse * 10 + digit
- number = number // 10
- print(reverse)

- OUTPUT:
- Enter a number: 123
- 321

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21. Palindrome Number Check

Question: Check if a number is a palindrome. **Explanation:** Compare number with its reverse. - **Input:** 121 - **Output:** Palindrome

- number = int(input("Enter a number: "))
- temp = number
- reverse = 0
- while temp > 0:
- digit = temp % 10
- reverse = reverse * 10 + digit
- temp = temp // 10
- if number == reverse:
- print("Palindrome")
- else:
- print("Not Palindrome")

- OUTPUT:
- Enter a number: 121
- Palindrome

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22. Count Vowels in String

Question: Count number of vowels in a string. **Explanation:** Loop and check for a, e, i, o, u. - **Input:** “apple” - **Output:** 2

- string = input("Enter a string: ")
- count = 0
- for ch in string:

- if ch.lower() in "aeiou":
- count = count + 1
- print(count)

- OUTPUT:
- Enter a string: apple
- 2

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23. Count Consonants in String

Question: Count consonants in a string. **Explanation:** Check for alphabetic characters not vowels. - **Input:** “apple” - **Output:** 3

- string = input("Enter a string: ")
- count = 0
- for ch in string:
- if ch.isalpha() and ch.lower() not in "aeiou":
- count = count + 1
- print(count)

- OUTPUT:
- Enter a string: apple
- 3

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24. Count Vowels and Consonants

Question: Count vowels and consonants in input string. **Explanation:** Maintain two counters. - **Input:** “apple” - **Output:** Vowels = 2, Consonants = 3

- string = input("Enter a string: ")
- vowels = 0
- consonants = 0
- for ch in string:
- if ch.isalpha():
- if ch.lower() in "aeiou":
- vowels = vowels + 1
- else:

- consonants = consonants + 1
- print("Vowels =", vowels)
- print("Consonants =", consonants)

- OUTPUT:
- Enter a string: apple
- Vowels = 2
- Consonants = 3

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25. Perfect Number Check

Question: Check if a number is perfect. **Explanation:** Sum of proper divisors equals the number. - **Input:** 28 - **Output:** Perfect number

- number = int(input("Enter a number: "))
- sum = 0
- for i in range(1, number):
- if number % i == 0:
- sum = sum + i
- if sum == number:
- print("Perfect number")
- else:
- print("Not Perfect number")

- OUTPUT:
- Enter a number: 28
- Perfect number

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26. Neon Number Check

Question: Check if a number is a neon number. **Explanation:** Square the number, sum digits, match original. - **Input:** 9 - **Output:** Neon number

- number = int(input("Enter a number: "))
- square = number * number
- sum = 0
- while square > 0:

- digit = square % 10
- sum = sum + digit
- square = square // 10
- if sum == number:
- print("Neon number")
- else:
- print("Not Neon number")

- OUTPUT:
- Enter a number: 9
- Neon number

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27. Strong Number Check

Question: Check if a number is a strong number. **Explanation:** Sum of factorial of digits equals the number. - **Input:** 145 - **Output:** Strong number

- number = int(input("Enter a number: "))
- temp = number
- sum = 0
- while temp > 0:
- digit = temp % 10
- factorial = 1
- for i in range(1, digit + 1):
- factorial = factorial * i
- sum = sum + factorial
- temp = temp // 10
- if sum == number:
- print("Strong number")
- else:
- print("Not Strong number")

- OUTPUT:
- Enter a number: 145
- Strong number

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28. Harshad Number Check

Question: Check if a number is divisible by the sum of its digits. **Explanation:** Calculate digit sum and check divisibility. - **Input:** 18 - **Output:** Harshad number

- number = int(input("Enter a number: "))
- temp = number
- sum = 0
- while temp > 0:
- digit = temp % 10
- sum = sum + digit
- temp = temp // 10
- if number % sum == 0:
- print("Harshad number")
- else:
- print("Not Harshad number")

- OUTPUT:
- Enter a number: 18
- Harshad number

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29. Fibonacci Series

Question: Print the Fibonacci series up to n terms. **Explanation:** Start with 0, 1 and continue with sum of last two. - **Input:** n = 5 - **Output:** 0 1 1 2 3

- n = int(input("Enter number of terms: "))
- first = 0
- second = 1
- for i in range(n):
- print(first, end=" ")
- next = first + second
- first = second
- second = next

- OUTPUT:
- Enter number of terms: 5
- 0 1 1 2 3

30. Check for Neon Number (Repeated)

Question: Again, check for a neon number (example). **Explanation:** Square number and sum digits. - **Input:** 9 - **Output:** Neon number

- number = int(input("Enter a number: "))
- square = number * number
- sum = 0
- while square > 0:
- digit = square % 10
- sum = sum + digit
- square = square // 10
- if sum == number:
- print("Neon number")
- else:
- print("Not Neon number")

- OUTPUT:
- Enter a number: 9
- Neon number